Deforestation Analysis Team Meeting Brief

Subnational democracy and forest loss in Colombia

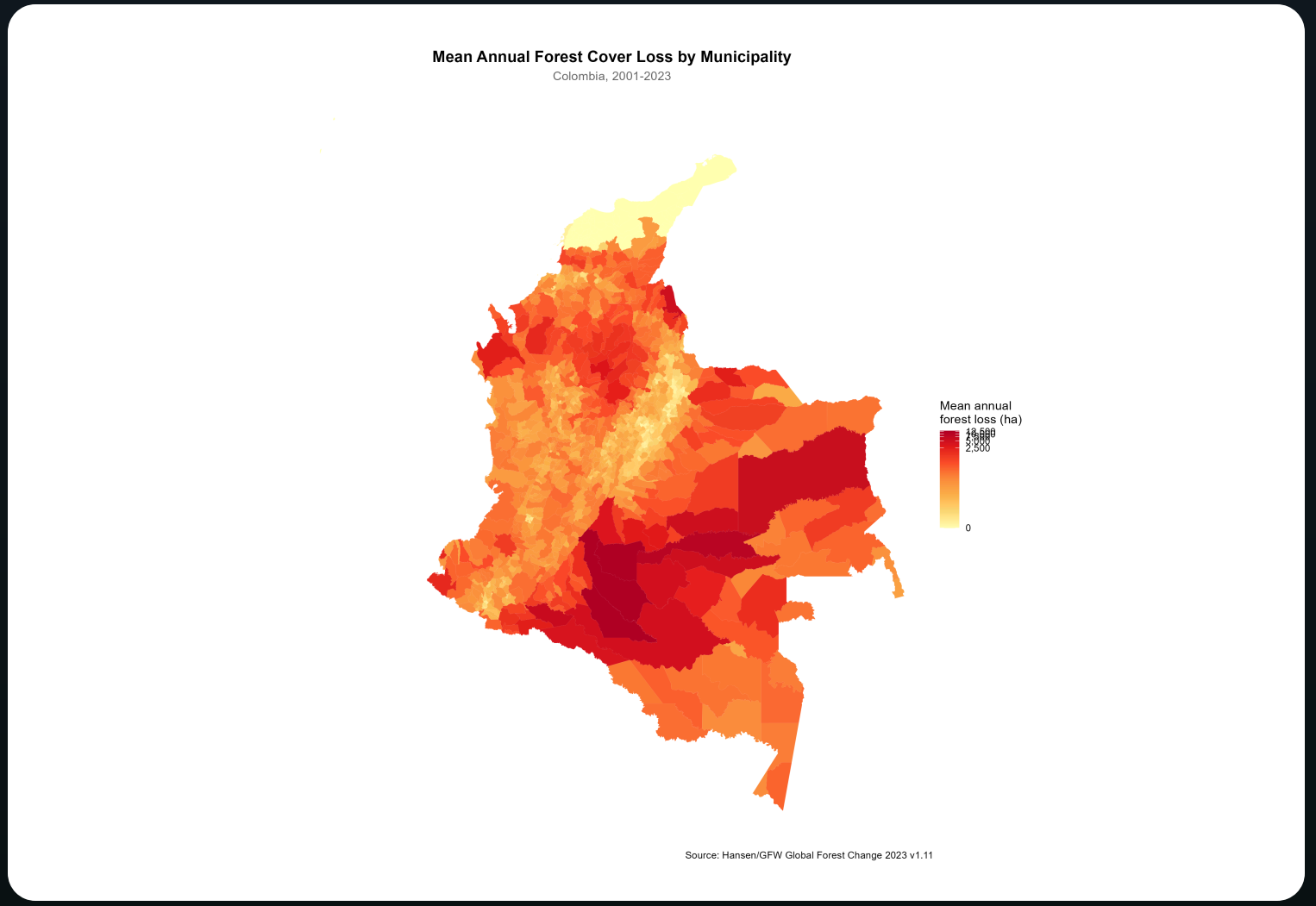
Based on the 2026 deforestation handoff memo. This deck focuses on the data, validation, and the institutional interpretation of the findings.

Prepared for: SN V-Dem team meeting

Location: **scripts/deforestation**

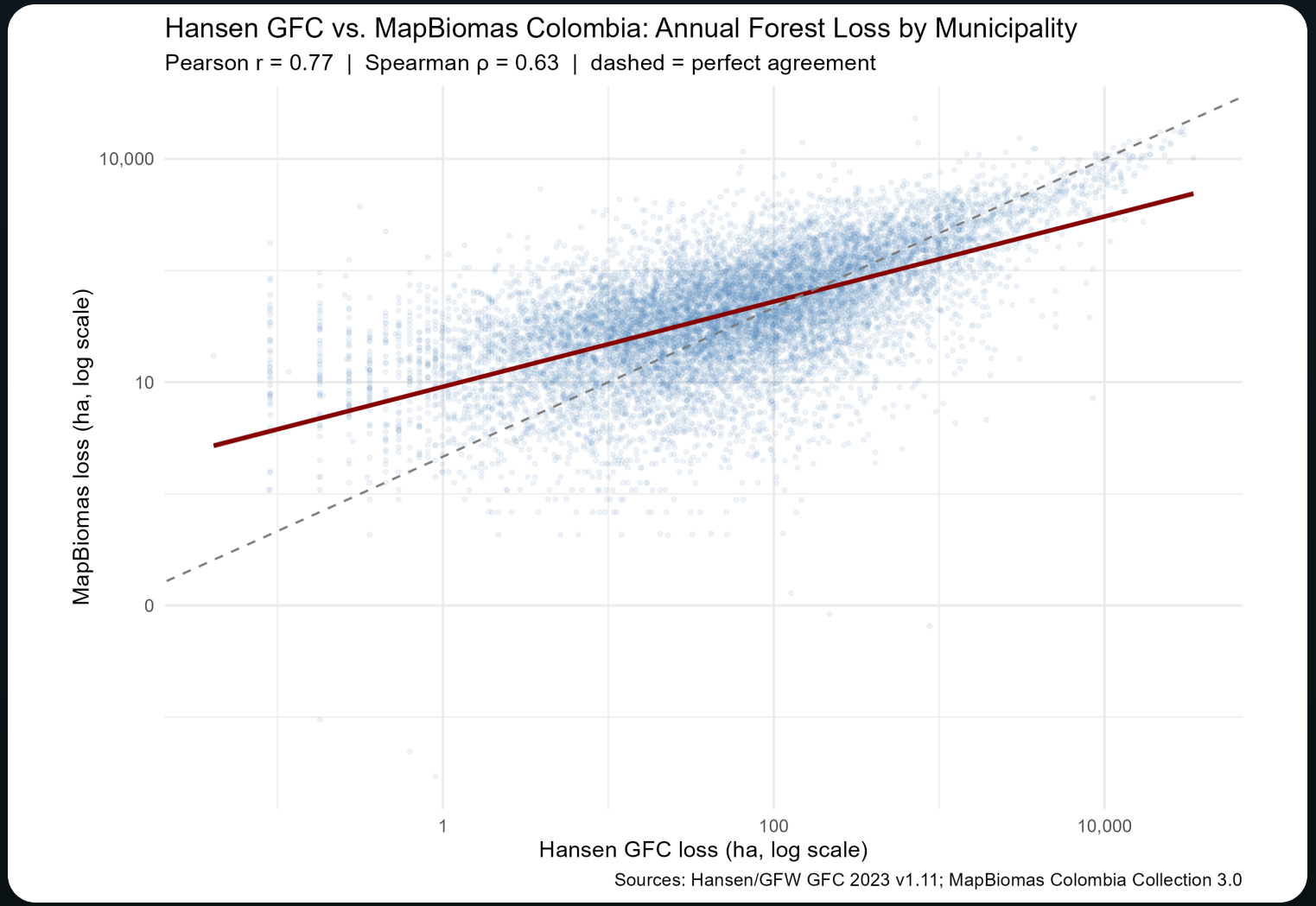
Research question

- Can subnational democracy explain deforestation patterns in Colombia?
- Does MapBiomas capture institutionally-relevant loss better than Hansen GFC?
- Does snelect show a non-linear frontier relationship?



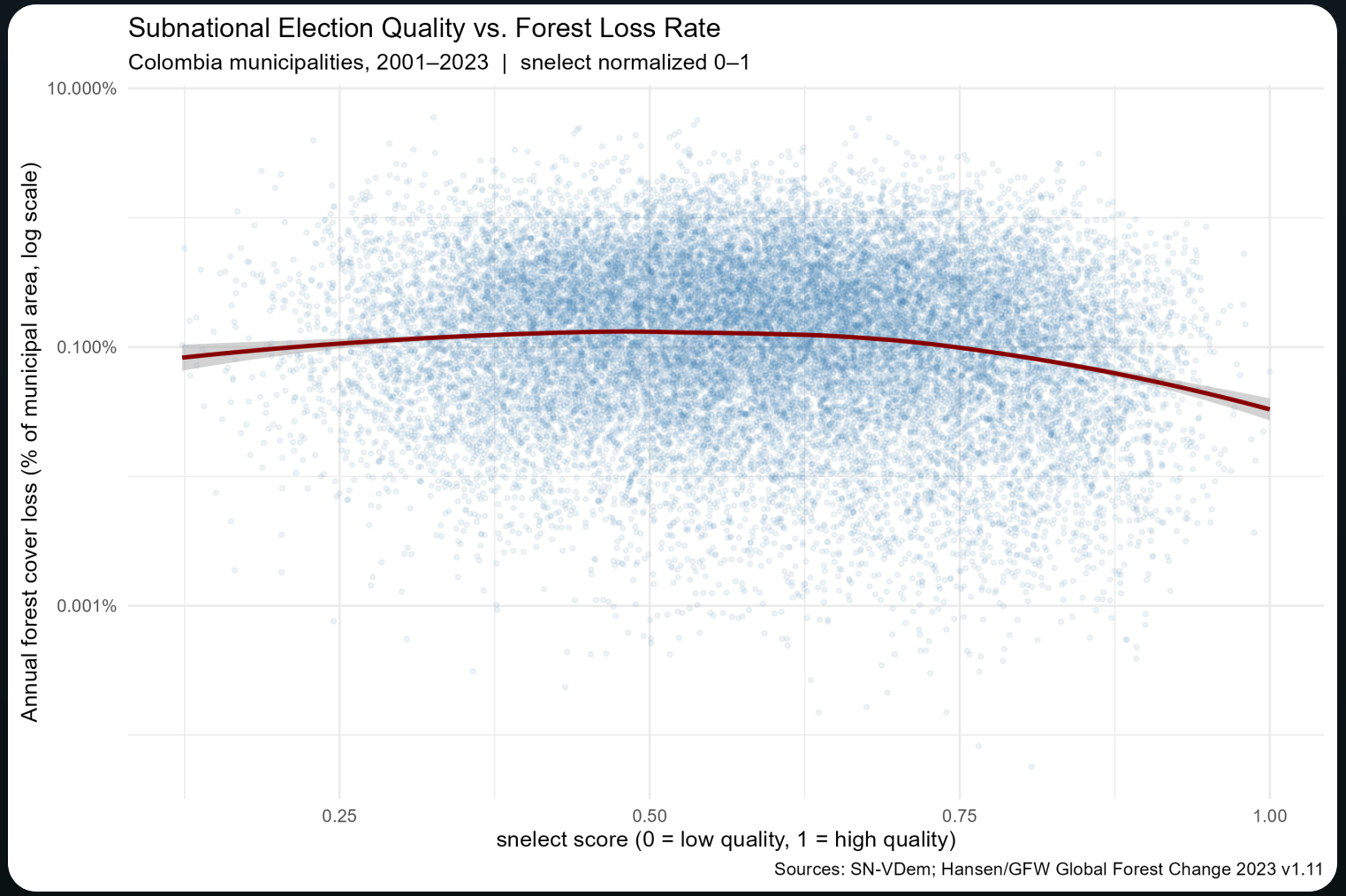
Data & validation

- Municipality-year panel for Colombia, 2001–2023.
- Democracy measure: **snelect** (elections component).
- Forest loss sources: Hansen GFC and MapBiomas.
- Cross-source validation: Pearson r = 0.77, Spearman ρ = 0.63.



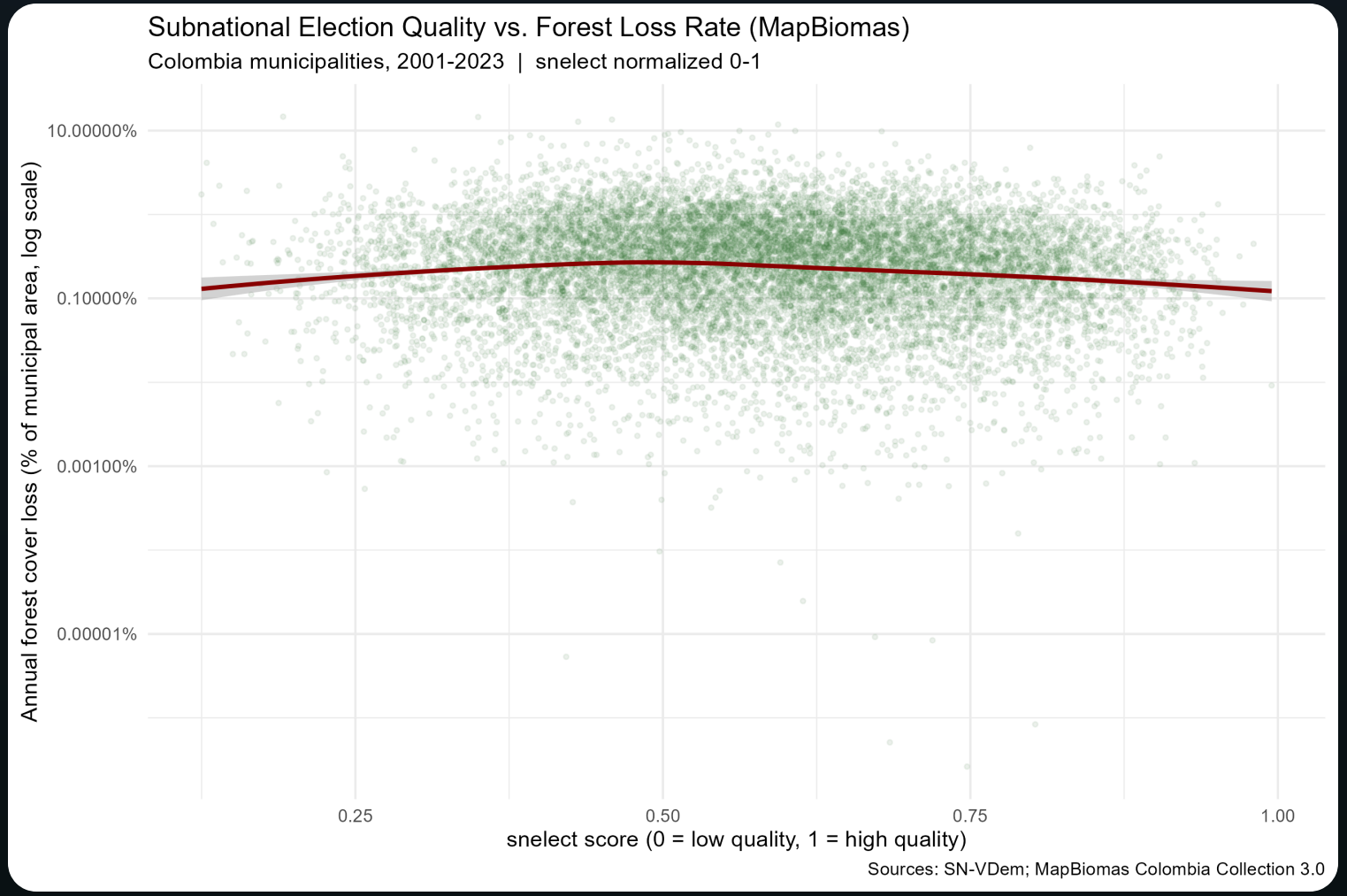
Main finding

- Non-linear U-shaped relationship.
- Very low snelect: low loss (remote, weak frontier pressure).
- Mid snelect: peak loss (frontier zones, weak institutions).
- High snelect: lower loss (stronger institutional capacity).



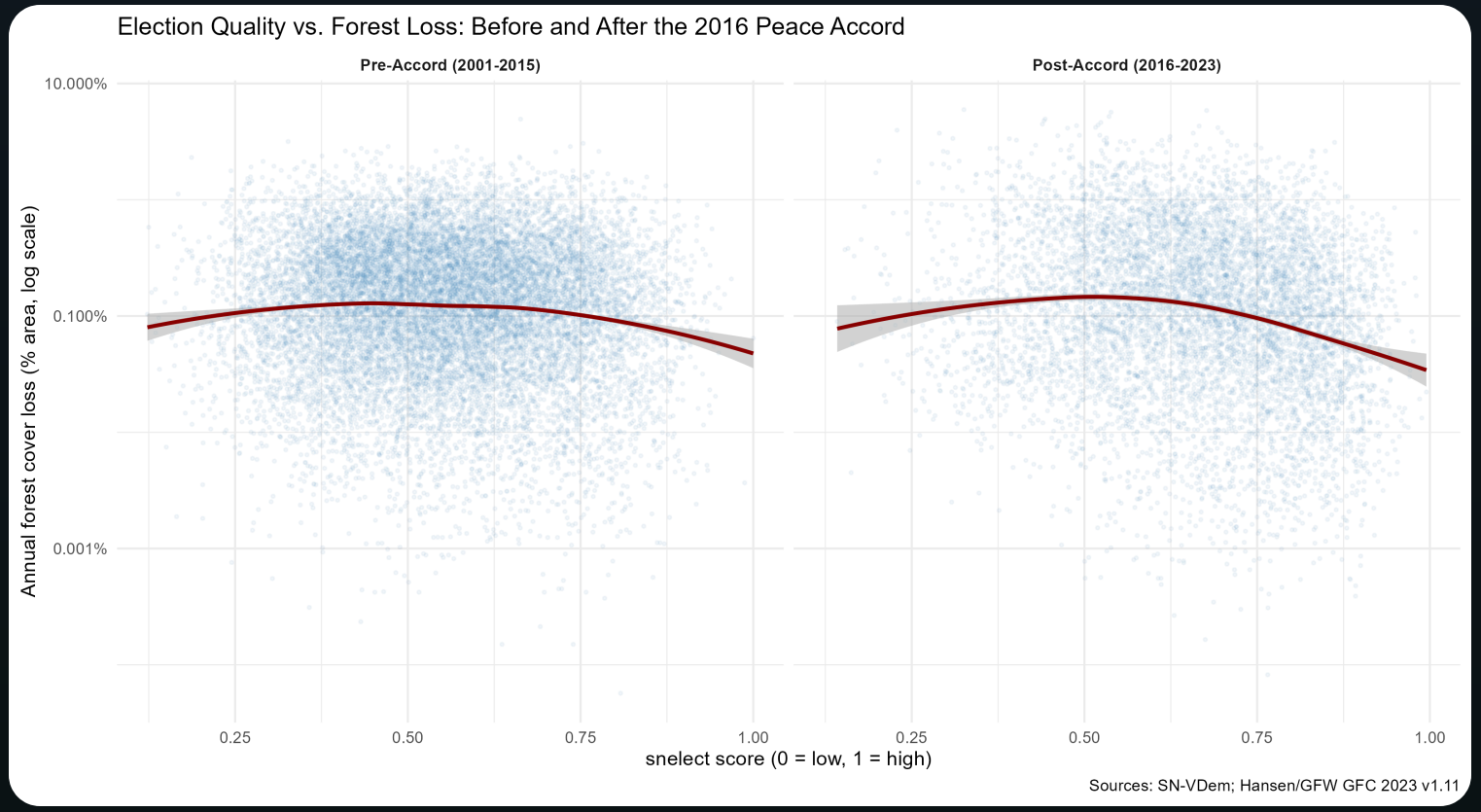
Robustness and mechanism

- MapBiomas captures land-use transitions, not all canopy disturbance.
- This makes it more relevant for institutional impact.
- Stronger correlation would support the democracy → land-use mechanism.



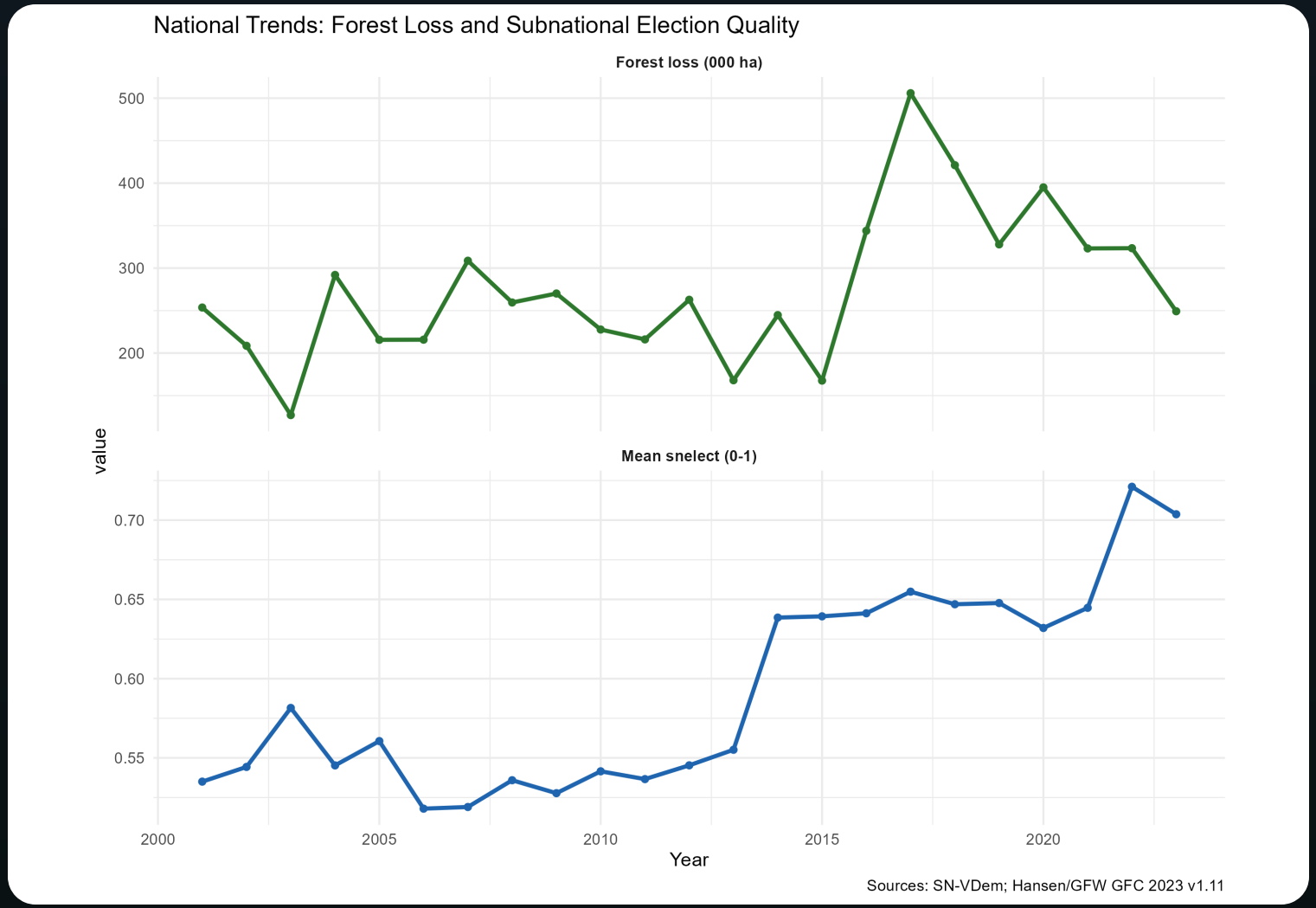
Temporal evidence

- Post-2016 Peace Accord surge is visible in the series.
- Former conflict zones opened to new clearing pressure.
- Subnational variation is essential to see this effect.



Global and conceptual fit

- Colombia is a hard test: functioning state, conservation law, conflict transition.
- National scores would miss within-country variation entirely.
- GFC and SN-VDem are globally available, so the pipeline is portable.



Recommendation

- Use this analysis as a proof of concept for subnational democracy research.
- Frame MapBiomas as the institutionally relevant loss measure.
- Highlight the U-shaped relationship and post-2016 forest loss surge.

